

# EFGP-SING E-step: GP-regression view plus derivation

## Big picture: there are two different E-step substitutions

There are two E-step moves:

**A. Fix  $q(x)$ , update  $q(f)$ .**

This replaces the **SING-GP inducing-point CAVI update**, specifically **Appendix K.3, Eq. (38)**. This is the line where the EFGP quantity  $h_r$  appears.

**B. Fix  $q(f)$ , update  $q(x)$ .**

This keeps the usual SING natural-gradient update, **Eq. (11)**, but changes how the transition-density expectations are computed. SING says those transition terms reduce to expectations of

$$f(x_i), \quad f(x_i)^\top \Sigma^{-1} f(x_i), \quad J_f(x_i)$$

under  $q(x_i)$ .

The key GP-regression translation is:

$q(f)$  update = uncertain-input heteroskedastic GP regression on latent velocities.

and

$q(x)$  update = SDE smoothing using GP posterior predictive mean, variance, and Jacobian.

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## 1. GP-regression view of the SING-GP drift update

### 1.1 Start from ordinary GP regression normal equations

Suppose one drift coordinate is represented in whitened EFGP features as

$$f_r(x) = \phi(x)^* w_r, \quad w_r \sim N(0, I).$$

For ordinary heteroskedastic Gaussian regression,

$$y_a = f_r(x_a) + \epsilon_a, \quad \epsilon_a \sim N(0, \lambda_a^{-1}),$$

the posterior over  $w_r$  is

$$q(w_r) = N(\mu_r, Q_r),$$

with

$$Q_r^{-1} = I + \Phi^* \Lambda \Phi,$$

and

$$Q_r^{-1} \mu_r = \Phi^* \Lambda y_r.$$

So, in GP language:

$$\boxed{Q_r^{-1} \mu_r \text{ is the weighted feature-response vector.}}$$

Equivalently,

$$\boxed{h_r := \Phi^* \Lambda y_r.}$$

That is the object we call  $h_r$ .

## 1.2 The SDE transition creates pseudo velocity observations

Euler says

$$x_{i+1} = x_i + \Delta_i f(x_i) + \epsilon_i, \quad \epsilon_i \sim N(0, \Delta_i \Sigma).$$

For coordinate  $r$ ,

$$x_{i+1,r} - x_{i,r} = \Delta_i f_r(x_i) + \epsilon_{i,r}, \quad \epsilon_{i,r} \sim N(0, \Delta_i \sigma_r^2).$$

Divide by  $\Delta_i$ :

$$\frac{x_{i+1,r} - x_{i,r}}{\Delta_i} = f_r(x_i) + \tilde{\epsilon}_{i,r}, \quad \tilde{\epsilon}_{i,r} \sim N\left(0, \frac{\sigma_r^2}{\Delta_i}\right).$$

So if the latent trajectory  $x_{0:T}$  were known, the drift update would be exactly GP regression with

$$\boxed{\text{input } x_i}$$

and

$$\boxed{\text{pseudo observation } y_{i,r}^{\text{pseudo}} = \frac{x_{i+1,r} - x_{i,r}}{\Delta_i}}$$

and precision weight

$$\boxed{\lambda_i = \frac{\Delta_i}{\sigma_r^2}.$$

Then the usual GP normal equations would be

$$Q_r^{-1} = I + \sum_i \frac{\Delta_i}{\sigma_r^2} \phi(x_i) \phi(x_i)^*,$$

and

$$Q_r^{-1} \mu_r = \sum_i \frac{\Delta_i}{\sigma_r^2} \phi(x_i) \left( \frac{x_{i+1,r} - x_{i,r}}{\Delta_i} \right).$$

The RHS simplifies to

$$Q_r^{-1} \mu_r = \frac{1}{\sigma_r^2} \sum_i \phi(x_i) (x_{i+1,r} - x_{i,r}).$$

This is the SDE-to-GP-regression bridge.

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### 1.3 With uncertain $q(x)$ , the pseudo regression problem is averaged over $q(x)$

In SING,  $x_i$  and  $x_{i+1}$  are not known. We have a Gaussian variational posterior

$$q(x_{0:T}).$$

So the exact  $q(f)$  update replaces the known-input regression quantities by expectations under  $q(x)$ :

$$Q_r^{-1} = I + \frac{1}{\sigma_r^2} \sum_i \Delta_i \mathbb{E}_q[\phi(x_i) \phi(x_i)^*],$$

and

$$Q_r^{-1} \mu_r = \frac{1}{\sigma_r^2} \sum_i \mathbb{E}_q[\phi(x_i) (x_{i+1,r} - x_{i,r})].$$

This is the cleanest form.

In this form, the RHS is literally

expected feature  $\times$  pseudo velocity response.

More specifically,

$$\mathbb{E}_q[\phi(x_i) (x_{i+1,r} - x_{i,r})]$$

is the uncertain-latent analogue of

$$\phi(x_i) (x_{i+1,r} - x_{i,r}).$$

So the RHS is the pseudo-observation term.

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## 1.4 Why the SING appendix writes the RHS in a two-term form

SING uses Stein's lemma to avoid integrating over the pair

$$q(x_i, x_{i+1}).$$

The key identity is

$$\mathbb{E}_q[\phi(x_i)(x_{i+1,r} - x_{i,r})] = \mathbb{E}_q[\phi(x_i)](m_{i+1,r} - m_{i,r}) + \mathbb{E}_q[J_\phi(x_i)]^*(S_{i,i+1} - S_i)_{\cdot r}.$$

So the RHS

$$Q_r^{-1}\mu_r = \frac{1}{\sigma_r^2} \sum_i \mathbb{E}_q[\phi(x_i)(x_{i+1,r} - x_{i,r})]$$

becomes

$$Q_r^{-1}\mu_r = \frac{1}{\sigma_r^2} \sum_i \{\mathbb{E}_q[\phi(x_i)](m_{i+1,r} - m_{i,r}) + \mathbb{E}_q[J_\phi(x_i)]^*(S_{i,i+1} - S_i)_{\cdot r}\}.$$

This is the same structure as SING-GP Appendix K.3, Eq. (38): one term involving expected kernel features times mean increments, plus one derivative/Stein correction involving  $S_{i,i+1} - S_i$ .

So the two-term SING expression is just an analytic rewrite of the pseudo-observation RHS.

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## 2. The SING-GP line we are replacing

### 2.1 Appendix K.3, Eq. (38)

SING-GP Appendix K.3 derives the sparse inducing-point update

$$S_u^{d*} = K_{zz} \left( K_{zz} + \Delta t \sum_i \mathbb{E}_{q(x_i)}[k_{zx_i} k_{x_i z}] \right)^{-1} K_{zz},$$

$$m_u^* = S_u^{d*} K_{zz}^{-1} \left[ \Delta t \sum_i \mathbb{E}_{q(x_i)}[k_{zx_i}] \left( \frac{m_{i+1} - m_i}{\Delta t} \right) + \Delta t \sum_i \mathbb{E}_{q(x_i)} \left[ \frac{dk_{zx}}{dx} \right] \left( \frac{S_{i,i+1} - S_i}{\Delta t} \right) \right].$$

This is Eq. (38). The paper then says the required kernel expectations are analytic for RBF kernels or approximable by quadrature, and that dependence on  $T$  enters only through sums over time.

We want to replace this sparse inducing-point update with an EFGP Fourier-weight update.

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### 3. Deriving the generic Gaussian feature update

#### 3.1 Set up a generic linear-in-weights drift

Ignore inducing points and Fourier features for one moment. Suppose one drift coordinate is represented as

$$f_r(x) = a(x)^\top \beta_r,$$

with Gaussian prior

$$\beta_r \sim N(0, P_0).$$

For the Euler transition,

$$x_{i+1,r} = x_{i,r} + \Delta_i f_r(x_i) + \varepsilon_{i,r}, \quad \varepsilon_{i,r} \sim N(0, \Delta_i \sigma_r^2),$$

the transition log-likelihood contribution involving  $f_r$  is

$$-\frac{1}{2\Delta_i \sigma_r^2} (x_{i+1,r} - x_{i,r} - \Delta_i f_r(x_i))^2.$$

Expand the square:

$$-\frac{1}{2\Delta_i \sigma_r^2} (x_{i+1,r} - x_{i,r})^2 + \frac{1}{\sigma_r^2} (x_{i+1,r} - x_{i,r}) f_r(x_i) - \frac{\Delta_i}{2\sigma_r^2} f_r(x_i)^2.$$

The first term does not involve  $\beta_r$ , so for updating  $q(\beta_r)$ , we only care about

$$\frac{1}{\sigma_r^2} \mathbb{E}_q[(x_{i+1,r} - x_{i,r}) f_r(x_i)] - \frac{\Delta_i}{2\sigma_r^2} \mathbb{E}_q[f_r(x_i)^2].$$

Plug in

$$f_r(x) = a(x)^\top \beta_r.$$

The quadratic term is

$$\mathbb{E}_q[f_r(x_i)^2] = \beta_r^\top \mathbb{E}_q[a(x_i) a(x_i)^\top] \beta_r.$$

The linear term is

$$\mathbb{E}_q[(x_{i+1,r} - x_{i,r}) f_r(x_i)] = \beta_r^\top \mathbb{E}_q[a(x_i) (x_{i+1,r} - x_{i,r})].$$

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### 3.2 Use Stein's lemma to avoid the pairwise expectation

For a vector function  $a(x_i)$  and jointly Gaussian pair  $(x_i, x_{i+1})$ ,

$$\mathbb{E}[a(x_i)(x_{i+1,r} - x_{i,r})] = \mathbb{E}[a(x_i)](m_{i+1,r} - m_{i,r}) + \mathbb{E}[J_a(x_i)]^\top (S_{i,i+1} - S_i)_{\cdot r}.$$

This is the source of the two bracketed RHS terms in SING-GP Eq. (38).

Define

$$d_i = m_{i+1} - m_i,$$

and

$$C_i = S_{i,i+1} - S_i.$$

Then the variational log-density for  $\beta_r$  has canonical Gaussian form

$$\log q^*(\beta_r) = -\frac{1}{2}\beta_r^\top \left[ P_0^{-1} + \frac{1}{\sigma_r^2} \sum_i \Delta_i \mathbb{E}[a(x_i)a(x_i)^\top] \right] \beta_r + \beta_r^\top \left[ \frac{1}{\sigma_r^2} \sum_i \{ \mathbb{E}[a(x_i)]d_{i,r} + \mathbb{E}[J_a(x_i)]^\top C_{i,r} \} \right].$$

So generically,

$$q^*(\beta_r) = N(\mu_r, Q_r),$$

with

$$Q_r^{-1} = P_0^{-1} + \frac{1}{\sigma_r^2} \sum_i \Delta_i \mathbb{E}[a(x_i)a(x_i)^\top],$$

and

$$Q_r^{-1} \mu_r = \frac{1}{\sigma_r^2} \sum_i \{ \mathbb{E}[a(x_i)]d_{i,r} + \mathbb{E}[J_a(x_i)]^\top C_{i,r} \}.$$

This is the derivation hiding behind the sparse-GP Eq. (38).

## 4. Recovering SING-GP Eq. (38) as a special case

In sparse GP notation,

$$u_r = f_r(z),$$

$$u_r \sim N(0, K_{zz}),$$

and

$$f_r(x) \approx k_{xz} K_{zz}^{-1} u_r.$$

So in the generic feature notation,

$$\beta_r = u_r,$$

$$P_0 = K_{zz},$$

and

$$a(x) = K_{zz}^{-1} k_{zx}.$$

Then

$$P_0^{-1} = K_{zz}^{-1}.$$

The precision update becomes

$$Q_u^{-1} = K_{zz}^{-1} + \frac{1}{\sigma_r^2} \sum_i \Delta_i K_{zz}^{-1} \mathbb{E}[k_{zx_i} k_{x_i z}] K_{zz}^{-1}.$$

If you invert this with the usual matrix identities, you get the SING-GP covariance expression

$$S_u^{d*} = K_{zz} \left( K_{zz} + \Delta t \sum_i \mathbb{E}[k_{zx_i} k_{x_i z}] \right)^{-1} K_{zz},$$

up to the  $\sigma_r^2$  scaling, which the paper suppresses or absorbs depending on notation.

The RHS is

$$Q_u^{-1} m_u = \frac{1}{\sigma_r^2} \sum_i \left\{ K_{zz}^{-1} \mathbb{E}[k_{zx_i}] d_{i,r} + K_{zz}^{-1} \mathbb{E} \left[ \frac{dk_{zx}}{dx} \right] C_{i,r} \right\}.$$

Multiplying by  $S_u$ , this becomes

$$m_u = S_u K_{zz}^{-1} \left[ \sum_i \mathbb{E}[k_{zx_i}] d_{i,r} + \sum_i \mathbb{E} \left[ \frac{dk_{zx}}{dx} \right] C_{i,r} \right],$$

which is exactly the bracketed term in Eq. (38).

So the SING-GP bracket is nothing mysterious: it is the **linear natural parameter** of the Gaussian posterior over the drift representation.

## 5. EFGP replacement for the $q(f)$ update

### 5.1 Fourier-weight representation

For EFGP, represent one drift coordinate as

$$f_r(x) = \phi(x)^* w_r,$$

where

$$w_r \sim N(0, I),$$

and

$$\phi(x) = D_\theta F_x.$$

Here

$$F_x[k] = \exp(2\pi i x^\top \xi_k),$$

and  $D_\theta$  is diagonal with square-root spectral weights.

So compared with the generic derivation,

$$\beta_r = w_r,$$

$$P_0 = I,$$

and

$$a(x) = \phi(x).$$

Therefore,

$$Q_r^{-1} = I + \frac{1}{\sigma_r^2} \sum_i \Delta_i \mathbb{E}_{q(x_i)} [\phi(x_i) \phi(x_i)^*],$$

and

$$Q_r^{-1} \mu_r = \frac{1}{\sigma_r^2} \sum_i \{ \mathbb{E}[\phi(x_i)] d_{i,r} + \mathbb{E}[J_\phi(x_i)]^* C_{i,r} \}.$$

Now define

$$A_r := Q_r^{-1},$$

and

$$h_r := \frac{1}{\sigma_r^2} \sum_i \{ \mathbb{E}[\phi(x_i)] d_{i,r} + \mathbb{E}[J_\phi(x_i)]^* C_{i,r} \}.$$

Then

$$A_r \mu_r = h_r.$$

That is why  $h_r$  replaces the bracketed SING-GP term: both are the linear natural parameter in the Gaussian posterior update. The difference is only the coordinate system:

$$\text{sparse GP: } u_r \sim N(0, K_{zz}),$$



$$\text{EFGP: } w_r \sim N(0, I).$$

Because EFGP uses whitened Fourier weights, the prior contribution is  $I$ , and we no longer need the outer factors  $S_u K_{zz}^{-1}$ .

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## 5.2 Monte Carlo approximation that preserves Toeplitz structure

The exact EFGP update contains

$$\mathbb{E}_{q(x_i)}[\phi(x_i)\phi(x_i)^*].$$

We approximate it by sampling

$$x_i^{(s)} \sim q(x_i), \quad s = 1, \dots, S.$$

Then

$$\mathbb{E}[\phi(x_i)\phi(x_i)^*] \approx \frac{1}{S} \sum_{s=1}^S \phi(x_i^{(s)})\phi(x_i^{(s)})^*.$$

Let

$$X = \{x_i^{(s)}\}_{i,s}.$$

Then

$$A_r \approx I + D_\theta [F_X^* W_r F_X] D_\theta,$$

where

$$(W_r)_{i,s} = \frac{\Delta_i}{S\sigma_r^2}.$$

The entries of the middle matrix are

$$(F_X^* W_r F_X)_{kl} = \sum_{i,s} \frac{\Delta_i}{S\sigma_r^2} \exp\left(2\pi i (x_i^{(s)})^\top (\xi_\ell - \xi_k)\right).$$

This depends only on the frequency difference  $\xi_\ell - \xi_k$ , so it is Toeplitz/BTTB on the equispaced Fourier grid. This is the same EFGP feature-space primitive:

$$(I + \Phi^* \Delta \Phi)u = \Phi^* z,$$

whose CG iterations use Toeplitz/BTTB matvecs at  $O(m^d \log m^d)$ , with only setup costs scaling linearly in  $n$ .

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## 5.3 Deriving the sampled RHS $h_r$

The first part of  $h_r$  is

$$h_{1,r} = \frac{1}{\sigma_r^2} \sum_i \mathbb{E}[\phi(x_i)] d_{i,r}.$$

Monte Carlo gives

$$h_{1,r} \approx \frac{1}{\sigma_r^2} \sum_i \frac{1}{S} \sum_s \phi(x_i^{(s)}) d_{i,r}.$$

Since

$$\phi = D_\theta F,$$

we get

$$h_{1,r} = D_\theta F_X^* a_r,$$

with

$$(a_r)_{i,s} = \frac{d_{i,r}}{S \sigma_r^2}.$$

The second part is the Stein correction:

$$h_{2,r} = \frac{1}{\sigma_r^2} \sum_i \mathbb{E}[J_\phi(x_i)]^* C_{i,r}.$$

For Fourier features,

$$\frac{\partial \phi_k(x)}{\partial x_j} = 2\pi i \xi_{k,j} \phi_k(x).$$

Therefore,

$$h_{2,r} \approx \sum_{j=1}^{D_x} (2\pi i \xi_j) \odot D_\theta F_X^* c_{j,r},$$

where

$$(c_{j,r})_{i,s} = \frac{C_{i,j,r}}{S \sigma_r^2}.$$

So

$$h_r = h_{1,r} + h_{2,r}.$$

This is the precise EFGP version of the bracketed term in SING-GP Eq. (38).

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## 6. Pair sampling and the pseudo-observation view

### 6.1 Pair sampling makes the pseudo-regression view completely literal

If we sample transition pairs

$$(x_i^{(s)}, x_{i+1}^{(s)}) \sim q(x_i, x_{i+1}),$$

then define pseudo observations

$$y_{i,r}^{(s)} = \frac{x_{i+1,r}^{(s)} - x_{i,r}^{(s)}}{\Delta_i}.$$

Then the sampled GP regression problem is:

inputs:

$$x_i^{(s)}$$

responses:

$$y_{i,r}^{(s)}$$

weights:

$$\lambda_{i,s} = \frac{\Delta_i}{S\sigma_r^2}.$$

The posterior precision is

$$A_r = I + \sum_{i,s} \lambda_{i,s} \phi(x_i^{(s)}) \phi(x_i^{(s)})^*,$$

and the RHS is

$$h_r = \sum_{i,s} \lambda_{i,s} \phi(x_i^{(s)}) y_{i,r}^{(s)}.$$

Substitute  $y_{i,r}^{(s)}$ :

$$h_r = \sum_{i,s} \frac{\Delta_i}{S\sigma_r^2} \phi(x_i^{(s)}) \left( \frac{x_{i+1,r}^{(s)} - x_{i,r}^{(s)}}{\Delta_i} \right),$$

so

$$h_r = \frac{1}{S\sigma_r^2} \sum_{i,s} \phi(x_i^{(s)})(x_{i+1,r}^{(s)} - x_{i,r}^{(s)}).$$

This is exactly a Monte Carlo estimator of

$$\frac{1}{\sigma_r^2} \sum_i \mathbb{E}_q[\phi(x_i)(x_{i+1,r} - x_{i,r})].$$

So pair sampling gives the cleanest conceptual story:

Update  $q(f_r)$  by GP regression on sampled latent velocities.

## 6.2 Why marginal sampling plus Stein may still be preferable computationally

Pair sampling is conceptually clean, but marginal sampling plus Stein gives the same target with potentially lower variance.

Pair-sampling estimator:

$$h_r \approx \frac{1}{S\sigma_r^2} \sum_{i,s} \phi(x_i^{(s)})(x_{i+1,r}^{(s)} - x_{i,r}^{(s)}).$$

Stein/marginal estimator:

$$h_r \approx \frac{1}{\sigma_r^2} \sum_i \left\{ \frac{1}{S} \sum_s \phi(x_i^{(s)})(m_{i+1,r} - m_{i,r}) + \frac{1}{S} \sum_s J_\phi(x_i^{(s)})^* (S_{i,i+1} - S_i)_{\cdot r} \right\}.$$

Both estimate the same quantity. The second avoids sampling the noisy increment  $x_{i+1} - x_i$ ; it uses the exact covariance correction  $S_{i,i+1} - S_i$ .

For EFGP, the derivative term is cheap because

$$\frac{\partial \phi_k(x)}{\partial x_j} = 2\pi i \xi_{k,j} \phi_k(x).$$

So marginal sampling plus Stein is probably the better estimator. But pair sampling is the best way to understand the update.

## 7. EFGP makes the pseudo-regression scalable

Once we have sampled pseudo inputs  $X$ , EFGP sees an ordinary heteroskedastic GP regression problem.

The normal equations are

$$A_r \mu_r = h_r,$$

with

$$A_r = I + D_\theta F_X^* W_r F_X D_\theta,$$

and

$$h_r = D_\theta F_X^* W_r y_r^{\text{pseudo}}.$$

This is exactly the same feature-space system as in EFGP heteroskedastic regression,

$$(I + \Phi^* \Delta \Phi) u = \Phi^* z,$$

where  $F^* \Delta F$  is Toeplitz/BTTB on an equispaced Fourier grid and can be applied by FFT, with only linear-in- $n$  setup and  $O(m^d \log m^d)$  CG iterations.

So the insider-GP translation is:

|                                                                                             |
|---------------------------------------------------------------------------------------------|
| SING-GP $q(f)$ update = uncertain-input heteroskedastic GP regression on latent velocities. |
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EFGP just supplies the fast solver for that regression problem.

## 8. Fix $q(f)$ , update $q(x)$

### 8.1 Where this plugs into SING

This is not Appendix K.3 Eq. (38). Eq. (38) only updates  $q(f)$ .

The  $q(x)$  update is the usual SING natural-gradient update, Eq. (11). That update requires gradients of the expected observation log-likelihood and the expected Euler transition log-density. The hard transition expectation is reduced by Proposition 1 to expectations involving

$$\mathbb{E}[f(x_i)],$$

$$\mathbb{E}[f(x_i)^\top \Sigma^{-1} f(x_i)],$$

and

$$\mathbb{E}[J_f(x_i)].$$

The paper says this reduction changes the expectation from one over  $q(x_i, x_{i+1})$  to one over  $q(x_i)$ , and that these expectations can be computed by quadrature or Monte Carlo.

So, for the  $q(x)$  update, we are subbing into the **three drift primitives** required by Proposition 1.

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## 8.2 Derive the $q(f)$ -integrated transition objective

Given  $q(f)$ , define

$$\bar{f}(x) := \mathbb{E}_{q(f)}[f(x)].$$

The Euler transition log-density is

$$\log \tilde{p}(x_{i+1} \mid x_i, f) = -\frac{1}{2\Delta_i} [x_{i+1} - x_i - \Delta_i f(x_i)]^\top \Sigma^{-1} [x_{i+1} - x_i - \Delta_i f(x_i)] + \text{const.}$$

Take expectation over  $q(f)$ . Let

$$\delta_i = x_{i+1} - x_i.$$

Then

$$\mathbb{E}_{q(f)}[\log \tilde{p}(x_{i+1} \mid x_i, f)] = -\frac{1}{2\Delta_i} \delta_i^\top \Sigma^{-1} \delta_i + \delta_i^\top \Sigma^{-1} \bar{f}(x_i) - \frac{\Delta_i}{2} \mathbb{E}_{q(f)}[f(x_i)^\top \Sigma^{-1} f(x_i)] + \text{const.}$$

So the random-GP drift only changes two objects:

$$f(x_i) \rightsquigarrow \bar{f}(x_i),$$

and

$$f(x_i)^\top \Sigma^{-1} f(x_i) \rightsquigarrow \mathbb{E}_{q(f)}[f(x_i)^\top \Sigma^{-1} f(x_i)].$$

The Stein correction in SING applies to the cross term

$$\delta_i^\top \Sigma^{-1} \bar{f}(x_i),$$

so the Jacobian primitive becomes

$$J_f(x_i) \rightsquigarrow J_{\bar{f}}(x_i).$$

That is the full logic of the substitution for the  $q(x)$  update.

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## 8.3 Compute these objects under EFGP

For each output coordinate,

$$q(w_r) = N(\mu_r, Q_r), \quad Q_r = A_r^{-1}.$$

The posterior mean drift is

$$\bar{f}_r(x) = \mathbb{E}[f_r(x)] = \phi(x)^* \mu_r.$$

The Jacobian is

$$\nabla_x \bar{f}_r(x) = \nabla_x \phi(x)^* \mu_r.$$

Using Fourier features,

$$\frac{\partial \bar{f}_r(x)}{\partial x_j} = \phi(x)^* [(2\pi i \xi_j) \odot \mu_r].$$

The second moment is

$$\mathbb{E}[f_r(x)^2] = \mathbb{E}[(\phi(x)^* w_r)^2] = (\phi(x)^* \mu_r)^2 + \phi(x)^* Q_r \phi(x).$$

For diagonal diffusion,

$$\mathbb{E}[f(x)^\top \Sigma^{-1} f(x)] = \sum_r \sigma_r^{-2} [(\phi(x)^* \mu_r)^2 + \phi(x)^* Q_r \phi(x)].$$

For full  $\Sigma$ , the mean part is

$$\bar{f}(x)^\top \Sigma^{-1} \bar{f}(x),$$

and the uncertainty correction is

$$\text{tr}(\Sigma^{-1} \text{Var}_{q(f)}[f(x)]).$$

If the output GPs are independent, this variance matrix is diagonal across  $r$ .

## 8.4 Why $\Phi Q_r \Phi^*$ appears

For a batch of evaluation points  $X$ ,

$$f_r(X) = \Phi_X w_r.$$

Since

$$q(w_r) = N(\mu_r, Q_r),$$

we have

$$q(f_r(X)) = N(\Phi_X \mu_r, \Phi_X Q_r \Phi_X^*).$$

Therefore,

$$\text{diag}(\Phi_X Q_r \Phi_X^*)$$

is the vector of posterior variances

$$\text{Var}_{q(f)}[f_r(x)]$$

at the evaluation points  $X$ .

We do not form  $Q_r$ . We only have

$$Q_r = A_r^{-1}.$$

So we estimate

$$\text{diag}(\Phi_X A_r^{-1} \Phi_X^*)$$

with Hutchinson probes:

$$\text{diag}(\Phi_X A_r^{-1} \Phi_X^*) \approx \frac{1}{J} \sum_{j=1}^J z_j \odot \Phi_X A_r^{-1} \Phi_X^* z_j.$$

Each probe uses:

$$z_j \rightarrow \Phi_X^* z_j \rightarrow A_r^{-1} \Phi_X^* z_j \rightarrow \Phi_X A_r^{-1} \Phi_X^* z_j.$$

That is one type-1 NUFFT, one CG solve, and one type-2 NUFFT. This is exactly the same matrix-free diagonal strategy used in EFGP for estimating  $\text{diag } \Sigma$ : rather than computing  $\Sigma_{ii} = e_i^\top \Sigma e_i$  directly, use  $z \odot \Sigma z$ , which costs  $O(J(n + m^d \log m^d))$ .

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## 9. Where each computation plugs in

### 9.1 $q(x)$ fixed, update $q(f)$

We replace **Appendix K.3 Eq. (38)**.

Sparse-GP object:

$$\sum_i \mathbb{E}[k_{zx_i} k_{x_i z}]$$

EFGP object:

$$D_\theta F_X^* W F_X D_\theta.$$

Sparse-GP object:



$$\sum_i \mathbb{E}[k_{zx_i}] d_{i,r}$$

EFGP object:

$$D_\theta F_X^* a_r.$$

Sparse-GP object:

$$\sum_i \mathbb{E} \left[ \frac{dk_{zx}}{dx} \right] C_{i,r}$$

EFGP object:

$$\sum_j (2\pi i \xi_j) \odot D_\theta F_X^* c_{j,r}.$$

Then solve:

$$A_r \mu_r = h_r.$$

That is the full replacement for  $S_u^*, m_u^*$ .

## 9.2 $q(f)$ fixed, update $q(x)$

We keep **SING Eq. (11)**, but when Eq. (11) asks for the transition expectation, Proposition 1 says the needed primitives are:

$$\mathbb{E}[f(x_i)], \quad \mathbb{E}[f(x_i)^\top \Sigma^{-1} f(x_i)], \quad \mathbb{E}[J_f(x_i)].$$

We provide these as:

$$\begin{aligned} \mathbb{E}[f(x_i)] &\leftarrow \bar{f}(x_i) = \Phi_i \mu. \\ \mathbb{E}[f(x_i)^\top \Sigma^{-1} f(x_i)] &\leftarrow \bar{f}(x_i)^\top \Sigma^{-1} \bar{f}(x_i) + \text{tr} [\Sigma^{-1} \text{Var}_{q(f)}(f(x_i))]. \\ \mathbb{E}[J_f(x_i)] &\leftarrow J_{\bar{f}}(x_i). \end{aligned}$$

These feed into the exact same SING natural-gradient update for  $h_i, J_i, L_i$ .

## 10. Algorithm block: E-step with EFGP

**Given:** current  $q(x_{0:T})$ , fixed kernel hyperparameters  $\theta$ , fixed diffusion  $\Sigma$ , fixed emissions, Fourier grid  $\{\xi_k\}_{k=1}^M$ , spectral weights  $D_\theta$ , MC sample count  $S$ , Hutchinson probe count  $J$ .

**Goal:** update  $q(f)$  and  $q(x)$  without forming  $\Phi$ ,  $\Phi^* W \Phi$ ,  $Q_r$ , or any dense covariance matrix.

### 1. Extract current SING moments.

For every transition  $i$ , get

$$m_i, \quad S_i, \quad S_{i,i+1}.$$

Define

$$d_i = m_{i+1} - m_i$$

and

$$C_i = S_{i,i+1} - S_i.$$

If inputs are present, replace

$$d_i \leftarrow m_{i+1} - m_i - \Delta_i B v_i.$$

### 2. Sample marginal latent locations for the $q(f)$ update.

Draw

$$x_i^{(s)} \sim N(m_i, S_i), \quad s = 1, \dots, S.$$

These are marginal samples, not transition-pair samples.

### 3. Build the EFGP Gram operator.

Use the sampled cloud  $X = \{x_i^{(s)}\}$  to build the BTTB generator for

$$F_X^* W_r F_X, \quad (W_r)_{i,s} = \frac{\Delta_i}{S \sigma_r^2}.$$

This replaces the Eq. (38) object

$$\sum_i \mathbb{E}[k_{zx_i} k_{x_i z}].$$

### 4. Define the Fourier-weight precision operator.

For each output dimension  $r$ ,

$$A_r v = v + D_\theta [F_X^* W_r F_X] D_\theta v.$$

### 5. Build the RHS $h_r$ .

Compute

$$h_{1,r} = D_\theta F_X^* a_r, \quad (a_r)_{i,s} = \frac{d_{i,r}}{S \sigma_r^2}.$$

Compute the derivative correction:

$$h_{2,r} = \sum_j (2\pi i \xi_j) \odot D_\theta F_X^* c_{j,r},$$

where

$$(c_{j,r})_{i,s} = \frac{C_{i,j,r}}{S\sigma_r^2}.$$

Set

$$h_r = h_{1,r} + h_{2,r}.$$

#### 6. Solve the EFGP $q(f)$ update.

Use CG:

$$A_r \mu_r = h_r.$$

Store

$$q(w_r) = N(\mu_r, A_r^{-1}),$$

but store  $A_r^{-1}$  only as a solve primitive.

#### 7. Evaluate EFGP drift moments for the $q(x)$ update.

At MC or quadrature points  $X_{\text{eval}} \sim q(x_i)$ , compute:

$$\bar{f}_r(X_{\text{eval}}) = \Phi_{\text{eval}} \mu_r.$$

Compute Jacobians using frequency-multiplied coefficients:

$$\partial_{x_j} \bar{f}_r(X_{\text{eval}}) = \Phi_{\text{eval}} [(2\pi i \xi_j) \odot \mu_r].$$

Estimate

$$\text{diag}(\Phi_{\text{eval}} A_r^{-1} \Phi_{\text{eval}}^*)$$

by Hutchinson probes.

#### 8. Substitute into the SING $q(x)$ update.

In the transition expectation from Proposition 1, use:

$$\begin{aligned} f(x_i) &\leftarrow \bar{f}(x_i), \\ f(x_i)^\top \Sigma^{-1} f(x_i) &\leftarrow \bar{f}(x_i)^\top \Sigma^{-1} \bar{f}(x_i) + \text{tr}[\Sigma^{-1} \text{Var}_{q(f)}(f(x_i))], \\ J_f(x_i) &\leftarrow J_{\bar{f}}(x_i). \end{aligned}$$

#### 9. Run the SING natural-gradient update for $q(x)$ .

Update the Gaussian Markov natural parameters

$$h_i, \quad J_i, \quad L_i$$

exactly as in SING Eq. (11), but using the EFGP drift moments above.

#### 10. Repeat inner E-step iterations if desired.

Alternate  $q(f)$  and  $q(x)$  updates while holding  $\theta, \Sigma, C, d$  fixed.

## 11. Complexity target

Let  $N$  be the number of transitions,  $M = m^d$ ,  $S$  the MC samples,  $k$  the number of CG iterations, and  $J$  the number of Hutchinson probes.

The  $q(f)$  update costs approximately

$$O(SN + M \log M) + O(kD_{\text{out}}M \log M).$$

The  $q(x)$  drift-moment evaluation costs approximately

$$O(S_{\text{eval}}N) + O(D_{\text{out}}D_{\text{lat}}M \log M) + O(J[kM \log M + S_{\text{eval}}N]).$$

So the desired structure is

$$\boxed{O(N) + f(M), \quad f(M) \approx O(kM \log M).}$$

The core implementation rule is:

$$\boxed{\text{Never form } \Phi, \quad \Phi^*W\Phi, \quad Q_r, \quad \text{or dense covariance matrices.}}$$

Everything should be implemented using type-1 NUFFT, type-2 NUFFT, BTTB matvecs, CG solves, Hutchinson probes, or SING's existing Gaussian-chain update.

## 12. Main takeaway

The reason  $h_r$  appears is that the  $q(f)$  update is just Gaussian conjugacy for a linear-in-weights drift model.

Sparse SING-GP uses

$$f_r(x) = k_{xz}K_{zz}^{-1}u_r, \quad u_r \sim N(0, K_{zz}),$$

so Eq. (38) is written in terms of  $S_u^*$ ,  $K_{zz}^{-1}$ , and the bracketed kernel RHS.

EFGP uses

$$f_r(x) = \phi(x)^* w_r, \quad w_r \sim N(0, I),$$

so the same Gaussian update becomes

$$A_r \mu_r = h_r.$$

The bracketed SING-GP term and  $h_r$  are the same object in different coordinates: the **linear natural parameter** of the Gaussian posterior over the drift representation.

Equivalently, in GP-regression language:

$$\boxed{h_r = \text{weighted feature-pseudo-response vector.}}$$

And the pseudo-response is the latent velocity

$$y_{i,r}^{\text{pseudo}} = \frac{x_{i+1,r} - x_{i,r}}{\Delta_i}.$$

With uncertain latents, SING either writes this as the Stein-corrected expectation

$$\mathbb{E}_q[\phi(x_i)](m_{i+1,r} - m_{i,r}) + \mathbb{E}_q[J_\phi(x_i)]^*(S_{i,i+1} - S_i)_r,$$

or we can estimate the same object by sampling transition pairs and regressing on sampled latent velocities.

So the whole E-step can be remembered as:

$$\boxed{q(f) \text{ update} = \text{EFGP regression on uncertain latent velocities.}}$$

$$\boxed{q(x) \text{ update} = \text{SING smoothing using EFGP posterior predictive moments.}}$$

M Step